

Chapter 2

Calculations used in Analytical Chemistry

First Week - Textbook Problems Assignment

Fundamentals of Analytical Chemistry, Skoog, et. al., 10th Ed., 2022

There are no problems assigned from Chapter 1. Chapter 2 is a review of general chemistry solution calculations: The problem list is below – remember, the problems with an “” have the answers in the back of the textbook, and the entire “solution” to these problems are located in your Student Solutions Manual.*

Chapter 2 Problems:

2-4, 2-7, 2-9, 2-11, 2-13, 2-14, 2-15, 2-16, 2-17, 2-18, 2-23, 2-25, 2-27, 2-29, 2-31, 2-33, 2-35, 2-37

Review of Some Basics from General Chemistry

SI units

Quantity	Unit	Symbol
length		
Mass		
Time		
Temperature		
Amount of substance		
Current		
Luminous intensity		

Prefix multipliers

Prefix	Symbol	Power of 10
Mega-		
Kilo-		
Centi-		
Milli-		
Micro-		
Nano-		
Pico-		

Masserini's Main Mandate: convert to moles the fundamental counting unit = 6.023×10^{23} atoms, ions, molecules, or anything

Formula weight: sum of the atomic weights of all the atoms present in the chemical formula of a species

Molecular weight: may be used if the chemical species is a molecule (two or more non-metals)

Millimole= is simply 1/1000 of a mole

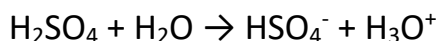
Molarity of a solution, symbolized by M, the number of moles of a solute contained in one liter of a solution

Molarity also equals the millimoles of solute contained in one milliliter of a solution

Analytical molarity, sometimes referred to as **formality**, F, specifies a recipe by which a solution is prepared.

Equilibrium molarity specifies the molar concentration of a particular species in a solution.

analytical molarity of H₂SO₄ in an aqueous 1.0 M solution is one mole H₂SO₄ /L, which corresponds to 98 g/L, since the formula weight of sulfuric acid is 98 g/mol. **equilibrium molarity** of H₂SO₄ in this same solution is 0.0 M, because sulfuric acid is classified as a strong acid and so is completely ionized in solution:



We'll use molarity M to express both analytical concentration and equilibrium concentration.

Concentrations are also expressed in terms of percentages. These include:

1. weight percent, (w/w%); weight of solute/weight of sol'n x 100%
2. volume percent, (v/v%); vol. of solute/vol. of sol'n x 100%
3. weight/volume percent, (w/v%); weight of solute **in g**/vol. sol'n **in mL** x 100%

Relative Concentrations are often used. 1 ppm = 1 molecule solute in one million molecules of air.

Another way of expressing concentration refers to parts per "something". For solutions, ppm refers to the weight of a solute, in mg per liter of solution, and ppb is weight of a solute in micrograms per liter of solution.

$$\frac{1 \text{ g}}{10^6 \text{ g sol'n}} \times \frac{1000 \text{ mg}}{1 \text{ g}} \times \frac{1 \text{ g sol'n}}{1 \text{ mL sol'n}} \times \frac{1000 \text{ mL}}{1 \text{ L}} = \frac{1 \text{ mg}}{\text{L}} = 1 \text{ ppm}$$

Another concentration term used is the p-function which is simply the negative log of the molar concentration of a species: $pX = -\log [X]$.

*Commercially prepared solutions, particularly acids and bases, are often described in terms of **specific gravity**.* This is the ratio of the mass of a certain substance to the mass of an equal volume of water at a given temperature. Specific gravity is unitless, **while density has units of mass per unit volume**, although the two terms can be used interchangeably, and will be in this course.

$$\frac{g \text{ sol'n}}{mL \text{ sol'n}}$$

Example Problem:

1. Describe the preparation of 500 mL of 6.00 M H_3PO_4 (F.W = 98.00) from the commercial reagent that is 85.00% (w/w) H_3PO_4 and has a specific gravity of 1.69.

If you know the molarity of the concentrated acid or base, there's a simple equation used to calculate how much of the concentrated reagent is necessary to make any amount of a dilute solution:

$$\underline{M}_{\text{conc}} \times V_{\text{conc.}} = \underline{M}_{\text{dil.}} \times V_{\text{dil.}}$$

$$\frac{\text{mol}_{\text{conc}}}{L} \times V_{\text{conc}}(L) = \frac{\text{mol}_{\text{dil}}}{L} \times V_{\text{dil}}(L)$$

$$\text{mol}_{\text{conc}} = \text{mol}_{\text{dil}}$$

2. The molarity of conc. nitric acid is 15 M describe the preparation of 250 mL of 3 M HNO₃

3. AlCl_3 (F.W. = 133.3) reacts with NaOH to produce $\text{Al}(\text{OH})_3$ and NaCl (58.44).
What volume, in mL, of 0.320 M NaOH will react with 2.320 g of AlCl_3 ? and b)
how many mg of NaCl will be formed?

4. A solution was prepared by dissolving 10.12 grams of $\text{KCl} \cdot \text{MgCl}_2 \cdot 6 \text{H}_2\text{O}$ (F.W. = 277.87) in sufficient water to give 2.000 L.

This an ionic salt (metal and non-metal) and so will totally dissociate:

Calculate a) the molar analytical concentration of $\text{KCl} \cdot \text{MgCl}_2 \cdot 6 \text{H}_2\text{O}$;

4b) calculate the molar concentration of Mg^{2+} .

c) calculate the molar concentration of Cl^- .

If you're ever unsure how to set up any equation, lay out the data that you have in such a way as to get the units that you're looking for at the end. In other words, do a unit analysis, and then pick up the calculator. Remember, the LAST thing that you do is pick up the calculator when problem solving.

d) Calculate the (w/v) percent of $\text{KCl} \cdot \text{MgCl}_2 \cdot 6 \text{H}_2\text{O}$.

e) Calculate the millimoles of Cl^- in 25.0 mL of this solution.

f) Calculate the ppm K^+ .

g) Calculate pMg for the solution.

h) Calculate pCl for the solution